# UWIRS x ESS Robotics Design Competition

**Proposal Stage Submission Template**

**For formative feedback**
**Grading will not be counted toward the final mark**

**Page Limits (Strict)**

- Cover Page: **1 page (not graded)**
- Proposal Content: **maximum 2 pages (graded)**
- References & Appendix: **not included in page count**

The page limit includes all text, figures, tables, and diagrams. Content beyond the limit may not be reviewed.

**Formatting Guidance (Recommended, not mandatory)**

Teams are encouraged to use a professional technical report style similar to IEEE conference papers for clarity and consistency.

Recommended:

- Two-column layout
- Times New Roman, 10–11 pt
- Single spacing
- Clear section headings
- Labelled figures and diagrams

Minor deviations will not be penalized; however, submissions must remain clear, concise, and professional.

Use clear headings and concise engineering writing.

**COVER PAGE (Not graded)**

Team Name
Team Members (Names and IDs)
Selected SDG
Proposed System Name
Course/Competition Name
Date

# PROPOSAL (Maximum 2 Pages)

## 1. Problem Definition and Context (30 marks)

### Problem Statement

Clearly describe the real-world problem, who is affected, and why the issue matters. Explain why a robotics-based solution is appropriate.

Write your response here.

### Engineering Constraints and Assumptions

Identify realistic constraints and assumptions (cost, environment, safety, power, size, regulations, etc.) and briefly justify each.

Write your response here.

### Measurable Goals

Define quantifiable performance targets.

Example format:

Metric | Target
Metric | Target
Metric | Target

## 2. Conceptual Robotics Approach (40 marks)

### Robot/Platform Selection and Justification

State the type of robot (wheeled, drone, manipulator, etc.) and justify the choice using engineering reasoning such as terrain, payload, energy, cost, or complexity.

Write your response here.

### System Architecture

Insert a labelled block diagram showing sensors, processing/controller, actuators, power system, and communication.

Briefly describe each subsystem and its function.

Write your explanation here.

### Feasibility

Provide simple estimates or rough calculations demonstrating that the solution is practical. Examples include time to complete tasks, energy usage, coverage area, speed, or accuracy.

Show reasoning or calculations here.

## 3. Sustainability and SDG Alignment (20 marks)

### SDG Relevance

Explain how the proposed system directly addresses the selected SDG target and contributes to solving the identified problem.

Write your response here.

### Environmental and Social Impact

Discuss environmental benefits, social/community benefits, and expected improvements. Quantify impacts where possible.

Write your response here.

### 4. Summary (Optional but Recommended)

Provide a brief 2–3 sentence summary highlighting the value and expected impact of your solution.

Write your response here.

# REFERENCES (Not counted in page limit)

Use IEEE citation format.

# APPENDIX (Optional, Not counted)

Optional diagrams, extra calculations, or supporting material.